

Does Cross-Attention Improve Audio–Text Emotion Recognition?

A Dual-Stream Fusion System on IEMOCAP, with an Evaluation-Protocol Leakage Audit

Speech and Language Processing — Project Report

Abisha Antony

[add remaining team members here]

Abstract

Speech emotion recognition systems intended for deployment must interpret utterances in which the words and the delivery convey different information. Text-only sentiment analysis cannot detect sarcasm or vocal distress, while audio-only recognition has no access to conversational content. Fusing the two modalities is the standard response, and cross-modal attention has become the default fusion mechanism in recent literature. However, a controlled study reported that cross-attention and self-attention perform comparably for audio–text fusion, and observed that its audio branch was the weakest of the three modalities evaluated. This raised the possibility that cross-attention had been handicapped by a weak acoustic representation rather than being intrinsically ineffective.

This project develops a dual-stream emotion recognition system that combines a published temporal acoustic encoder, extended with gammatone cepstral features, with a recurrent semantic encoder, and tests whether a stronger acoustic stream allows cross-attention to outperform self-attention. All experiments use the IEMOCAP corpus under leave-one-session-out cross-validation, which guarantees speaker-independent evaluation. The published acoustic baseline was reproduced before extension, achieving 92.08% on RAVDESS and 71.65% on IEMOCAP.

Fusion substantially outperformed both unimodal streams: 67.65% weighted accuracy against 60.15% for text alone and 53.86% for audio alone. However, the central hypothesis was not supported. Cross-attention reached 67.22% and self-attention 66.28%, a difference of 0.94 points at $p = 0.702$, and neither exceeded simple feature concatenation at 67.65%. A separate evaluation audit found that a randomised train–test split inflates accuracy by 19.65 to 20.47 points on RAVDESS and 7.96 points on IEMOCAP; the RAVDESS inflation was traced to duplicate recordings rather than speaker overlap. These results indicate that fusion of complementary modalities yields reliable gains, whereas the choice of attention mechanism does not, and that evaluation protocol can influence reported accuracy more than architectural choice.

Keywords: speech emotion recognition; multimodal fusion; cross-modal attention; IEMOCAP; gammatone cepstral coefficients; speaker-independent evaluation; data leakage; temporal convolutional networks

Contents

1. Introduction
2. Literature Review
3. Preliminary Concepts and Theoretical Background
4. Materials and Methods
5. Proposed Methodology and Implementation
6. Experimental Framework
7. Results and Discussion
8. Conclusion and Future Work
- References

List of Figures

- Figure 1. System architecture and information flow
Figure 2. Weighted accuracy by configuration
Figure 3. Evaluation-protocol inflation across corpora and models
Figure 4. Confusion matrix of the fusion model on the held-out session

List of Tables

- Table 1. Critical comparison of prior work
Table 2. Hardware and software environment
Table 3. Dataset description
Table 4. Label mapping and exclusions
Table 5. Per-session composition of the corpus
Table 6. Cross-validation fold structure
Table 7. Primary results across all configurations
Table 8. Cross-attention versus self-attention significance tests
Table 9. Ablation study
Table 10. Evaluation-protocol leakage
Table 11. Acoustic similarity between clip pairs in RAVDESS
Table 12. Cross-corpus generalisation
Table 13. Per-class performance of the fusion model

1. Introduction

1.1 Context and significance

Automatic emotion recognition from speech underpins a range of human-computer interaction systems. Call centres route distressed callers to senior operators; telehealth platforms monitor patients for indicators of depression or anxiety across repeated consultations; automotive systems detect driver stress and adapt the cabin environment; conversational assistants modulate their responses according to the user's affective state. In each case the system must infer an internal state from a signal that was never designed to transmit it explicitly.

1.2 The specific problem

Emotion is expressed through two channels that can diverge. The lexical channel carries what was said; the paralinguistic channel carries how it was said. When the two agree, either alone is sufficient. When they diverge — in sarcasm, in suppressed distress, in politeness masking anger — a system attending to only one channel will be confidently wrong.

The canonical illustration is an utterance such as "Oh, what a fantastic day" spoken through tears. A text-only sentiment classifier reads unambiguously positive lexical content and returns a positive label. An audio-only classifier detects the vocal correlates of distress but has no access to the semantic context that makes the utterance interpretable. Only a system with access to both can identify the contradiction and resolve it.

1.3 Existing approaches and their limitations

Three solution families dominate. Unimodal acoustic models extract spectral or cepstral representations and classify them with convolutional or recurrent architectures; they capture delivery but not content. Unimodal textual models apply sentiment analysis to transcripts; they capture content but are blind to delivery. Multimodal fusion models combine both, and have consistently outperformed either alone.

Within fusion, the field has converged on cross-modal attention, in which one modality attends over the other so that the network can dynamically weight whichever stream is more informative. Tensor-based and subspace-based alternatives exist but scale poorly or introduce tuning instability. Cross-attention is now adopted as a default in most recent work.

1.4 Research gap

The gap addressed here is one of evidence and evaluation rather than absence of work. Two specific problems are identified.

First, an unverified architectural default. A controlled comparison by Rajan, Brutti and Cavallaro (2022) found cross-attention and self-attention to be statistically comparable for audio-text fusion on IEMOCAP. Subsequent work has largely continued to adopt cross-attention without re-running that ablation on its own architecture. Their audio branch, however, achieved only 36.5% accuracy in isolation — the weakest of the three modalities they evaluated — which leaves open whether cross-attention failed intrinsically or was merely supplied with an uninformative stream to attend over. That confound has not been resolved.

Second, an evaluation-protocol problem. Reported accuracies in speech emotion recognition are frequently obtained under randomised cross-validation, which permits the same speaker, and in some

corpora the same recording session, to appear in both training and test partitions. The magnitude of the resulting inflation is rarely quantified, and consequently the deployment-relevant accuracy of published systems is unclear.

1.5 Objectives

1. Reproduce a published acoustic baseline on two corpora to establish that the experimental setup is sound before extending it.
2. Develop a dual-stream architecture combining a temporal acoustic encoder with a recurrent semantic encoder, and extend the acoustic representation with gammatone cepstral coefficients.
3. Evaluate cross-attention against self-attention under identical parameter budgets and identical data, and determine whether a stronger acoustic stream alters the published conclusion.
4. Quantify the contribution of each modality and of the gammatone extension through controlled ablation.
5. Measure the effect of evaluation protocol on reported accuracy by varying only the train–test partitioning while holding model, features and hyperparameters constant.
6. Implement an inference interface that operates on previously unseen audio through automatic speech recognition, and characterise the gap between benchmark and deployment conditions.

1.6 Contributions

- A dual-stream architecture in which cross-attention and self-attention variants are implemented with identical parameter counts (849,956), isolating the mechanism from model capacity (Chapter 5).
- An independent replication of the cross-attention null result under materially different conditions — four emotion classes rather than seven, a temporal convolutional acoustic encoder rather than a recurrent one, and an extended feature set (Section 7.3).
- A quantified measurement of evaluation-protocol inflation on two corpora, with a decomposition attributing the effect on RAVDESS to duplicate recordings rather than speaker overlap (Section 7.5).
- A cross-corpus generalisation measurement showing that the trained system retains only 35.0% accuracy when transferred to an unseen corpus (Section 7.6).
- A complete inference pipeline including automatic transcription, with a measured characterisation of the benchmark-to-deployment gap (Section 7.7).

1.7 Organisation of this report

Chapter 2 reviews prior work and derives the research gap. Chapter 3 introduces the signal-processing and neural concepts required. Chapter 4 documents materials and procedures. Chapter 5 presents the proposed architecture at module level with pseudocode. Chapter 6 defines the experimental framework, research questions and metrics. Chapter 7 reports and interprets results. Chapter 8 concludes and identifies future work.

2. Literature Review

2.1 Domain and terminology

Speech emotion recognition (SER) assigns an affective label to a spoken utterance. Categorical formulations predict discrete classes; dimensional formulations regress continuous valence, arousal and dominance. This project adopts the categorical four-class formulation standard on IEMOCAP. Multimodal emotion recognition extends SER by incorporating the lexical channel, and where both are present the term fusion denotes the mechanism by which the two representations are combined.

2.2 Benchmark corpora

Two corpora anchor most comparative work. Busso et al. (2008) introduced IEMOCAP, a dyadic corpus of ten actors recorded across five two-person sessions combining scripted and improvised material. It remains the standard testbed for audio–text fusion, though its size and single-laboratory recording conditions limit cross-corpus generalisation. Poria et al. (2019) introduced MELD, a multi-party television-dialogue corpus of approximately 13,000 utterances that better reflects naturalistic multi-speaker conversation but whose label distribution is heavily skewed toward the neutral class. Livingstone and Russo (2018) introduced RAVDESS, an acted corpus of 24 professional actors producing fixed carrier sentences under eight emotion labels; its controlled structure makes it convenient but, as Section 7.5 demonstrates, structurally unsuited to randomised evaluation.

2.3 Fusion architectures

Early fusion work modelled cross-modal interaction explicitly. Zadeh et al. (2017) proposed the Tensor Fusion Network, which captures unimodal, bimodal and trimodal interactions through a Cartesian product of modality representations; the formulation is expressive but parameter count grows exponentially with the number of modalities. Hazarika et al. (2020) proposed MISA, decomposing each modality into modality-invariant and modality-specific subspaces before fusion, which improves representation quality at the cost of additional loss terms and sensitivity to their weighting.

Tsai et al. (2019) introduced the Multimodal Transformer, applying bidirectional cross-modal attention across time steps to handle unaligned sequences without explicit alignment. It processes modality pairs rather than a joint three-way interaction, leaving trimodal dependencies weakly modelled. Dutta and Ganapathy (2023) narrowed the focus to audio–text with HCAM, a hierarchical cross-attention model attending over both utterance-level and dialogue-level context; extending the context window improves emotional grounding but trades directly against inference latency.

More recent work generally augments cross-attention rather than reconsidering it. MER-CAPF (2026) adds multi-granularity pooling and reports 74.7%, 63.3% and 67.4% on IEMOCAP, MELD and MOSEI respectively, though its pooling configuration is tuned per dataset with no demonstrated transfer. Li et al. (2025) pair cross-attention with contrastive learning and gain 3.4 and 5.7 weighted-F1 points on IEMOCAP and MELD, with improvement concentrated almost entirely in minority classes.

2.4 The cross-attention question

Against this trend, Rajan et al. (2022) conducted a direct controlled comparison of cross-attention against self-attention for multimodal emotion recognition on IEMOCAP. Both models improved on the prior state of the art, but their performance was statistically comparable; cross-attention conferred no reliable advantage. For the audio–text bimodal configuration specifically, self-attention was significantly better

(weighted accuracy 0.518 against 0.501). Their unimodal results are equally relevant: text alone reached 0.474, vision 0.454, and audio only 0.365 — the weakest modality by a substantial margin.

Unresolved confound. Because their audio branch was a comparatively simple convolutional–recurrent encoder and was the weakest of their three modalities, it cannot be determined from their results whether cross-attention is intrinsically no better than self-attention, or whether it was disadvantaged by an uninformative acoustic stream. This project resolves that confound by substituting a substantially stronger acoustic encoder while holding the fusion comparison otherwise constant.

2.5 Acoustic representation

Mel-frequency cepstral coefficients remain the dominant acoustic representation. Gammatone frequency cepstral coefficients, derived from a filterbank modelling cochlear response rather than a perceptual pitch scale, have been reported to be more robust under noise. Two 2025 studies support their complementary value: a Mel-scale gammatone filterbank formulation outperforms the conventional ERB-based variant on Emo-DB and RAVDESS, and a comparison of gammatone coefficients with a cubic SVM on CREMA-D and SAVEE finds refined GTCC narrows but does not close the gap to MFCC. Neither feature dominates alone, which motivates their combination.

Ye et al. (2023) proposed TIM-Net, a temporal-aware bidirectional multi-scale network that learns affective representations at multiple temporal resolutions using dilated causal convolutions. It reports 90.04/90.07 UAR/WAR on RAVDESS and is among the strongest published acoustic encoders with publicly available code and weights, making it suitable as the acoustic backbone here.

2.6 Comparison of prior work

Table 1 compares the studies most directly relevant to this project along technical dimensions rather than bibliographic ones. The final column identifies the limitation that this project either inherits or addresses.

Study	Method	Data	Key metric	Important limitation
Busso et al. (2008)	Corpus	IEMOCAP	10 speakers, 12 h	Single laboratory; limited cross-corpus transfer
Zadeh et al. (2017)	Tensor fusion	CMU-MOSI	Explicit n-modal interaction	Parameter count grows exponentially with modalities
Tsai et al. (2019)	Cross-modal transformer	CMU-MOSEI	Handles unaligned sequences	Pairwise only; weak three-way interaction
Hazarika et al. (2020)	Invariant/specific subspaces	MOSI, MOSEI	Improved representation quality	Multiple loss terms; sensitive to weighting
Rajan et al. (2022)	Cross- vs self-attention	IEMOCAP, 7-class	WA 0.518 (self) / 0.501 (cross)	Audio branch weakest modality (0.365); confound unresolved
Ye et al. (2023)	TIM-Net (dilated temporal)	RAVDESS + 5	UAR/WAR 90.04/90.07	Evaluated under 10-fold random CV
Dutta & Ganapathy (2023)	Hierarchical cross-attention	IEMOCAP, MELD	Dialogue-level context	Context window trades against latency

Study	Method	Data	Key metric	Important limitation
Li et al. (2025)	Cross-attn + contrastive	IEMOCAP, MELD	+3.4 / +5.7 weighted-F1	Gains concentrated in minority classes

Table 1. Critical comparison of prior work. Values are as reported in the original studies under their own evaluation settings and are not directly comparable across rows.

2.7 Recurring limitations and the resulting gap

Three limitations recur across this literature. Cross-attention is adopted as a default despite a controlled study finding no advantage, and that ablation is rarely repeated. Evaluation protocols vary between studies — randomised cross-validation in some, speaker- or session-independent partitioning in others — while reported accuracies are frequently compared as though they were commensurable. Finally, acoustic streams are commonly the weakest modality in fusion systems, yet the consequences of that weakness for the fusion mechanism itself are not examined.

This project addresses the intersection of the first and third: whether the published cross-attention null result persists when the acoustic stream is strengthened. In the course of doing so it also quantifies the second.

3. Preliminary Concepts and Theoretical Background

3.1 Source-filter model of speech

Speech production is modelled as an excitation source filtered by the vocal tract. The vocal folds generate a quasi-periodic source determined largely by pitch; the vocal tract acts as a time-varying filter shaping that source into distinguishable sounds. In the frequency domain the observed spectrum is the product of source and filter contributions:

$$S(f) = E(f) \cdot H(f) \quad (1)$$

where $S(f)$ is the observed spectrum, $E(f)$ the excitation and $H(f)$ the vocal-tract transfer function. Emotional expression is carried substantially by $H(f)$ — articulatory tension, breathiness, constriction — while $E(f)$ encodes speaker-specific pitch that is largely irrelevant to the task and actively harmful if memorised. Cepstral analysis separates the two: taking logarithms converts the product into a sum,

$$\log S(f) = \log E(f) + \log H(f) \quad (2)$$

after which a decorrelating transform concentrates the slowly varying filter component into low-order coefficients and the rapidly varying source component into high-order ones. Retaining only the low-order coefficients therefore discards pitch while preserving articulation.

3.2 Mel-frequency cepstral coefficients

MFCCs implement the above with a filterbank spaced on the mel scale, an empirical approximation of human pitch perception derived from psychoacoustic matching experiments:

$$m = 2595 \cdot \log_{10}(1 + f / 700) \quad (3)$$

where f is frequency in hertz and m the corresponding mel value. The power spectrum is passed through M triangular filters spaced uniformly in m , giving filter energies $E_1 \dots E_M$. The cepstral coefficients follow from the discrete cosine transform of their logarithms:

$$c(k) = \sum_{m=1}^M \log(E_m) \cdot \cos[k(m - 1/2) \pi / M], \quad k = 1 \dots K \quad (4)$$

where K is the number of retained coefficients. In this work $M = 128$ and $K = 39$.

3.3 Gammatone frequency cepstral coefficients

GFCCs follow the identical pipeline but replace the mel triangular filterbank with a gammatone filterbank. The gammatone filter is defined by the impulse response

$$g(t) = a \cdot t^{(n-1)} \cdot e^{(-2\pi b t)} \cdot \cos(2\pi f_c t + \phi) \quad (5)$$

where n is the filter order, b the bandwidth parameter, f_c the centre frequency, a an amplitude scaling term and ϕ the phase. This form was derived to fit measured impulse responses of auditory nerve fibres and therefore models the physical filtering performed by the basilar membrane, rather than approximating a perceptual judgement. Filters are spaced on the equivalent rectangular bandwidth scale:

$$ERB(f) = 24.7 \cdot (4.37 f / 1000 + 1) \quad (6)$$

Two properties matter for this application. Gammatone filters have smooth rather than piecewise-linear roll-off, so energy from a narrowband disturbance contaminates neighbouring bands less; and ERB spacing allocates finer resolution at low frequencies, where most speech energy resides. Both contribute to the improved noise robustness reported in the literature. In this work 48 gammatone filters are used and 13 coefficients retained.

3.4 Dilated causal convolution

A one-dimensional convolution is causal if the output at time t depends only on inputs at times $\leq t$, and dilated if it samples the input at a fixed stride d :

$$y(t) = \sum_{i=0}^{k-1} w(i) \cdot x(t - d \cdot i) \quad (7)$$

where k is kernel size, w the kernel weights and d the dilation rate. Stacking L layers with exponentially increasing dilation yields a receptive field that grows exponentially in depth while parameter count grows only linearly:

$$R = 1 + \sum_{l=1}^L (k - 1) \cdot 2^{l-1} \quad (8)$$

This property allows a comparatively shallow network to condition on an entire utterance, which is necessary because emotional cues are distributed across the utterance rather than localised.

3.5 Scaled dot-product and multi-head attention

Attention maps a set of queries against key-value pairs. Given query matrix Q , key matrix K and value matrix V with key dimension d_k :

$$\text{Attention}(Q, K, V) = \text{softmax}(Q K^T / \sqrt{d_k}) V \quad (9)$$

The scaling by $\sqrt{d_k}$ prevents the dot products from growing large enough to saturate the softmax. Multi-head attention applies h independent projections and concatenates the results, allowing different heads to attend to different relational patterns:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O, \quad \text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \quad (10)$$

The distinction central to this project. Self-attention sets Q , K and V from the same modality, so a stream refines its own representation using its internal structure. Cross-attention takes Q from one modality and K , V from the other, so one stream retrieves information from the other. The two are structurally identical apart from where Q , K and V originate — which is what permits a comparison at equal parameter count.

3.6 Evaluation metrics

Weighted accuracy (WA) is the proportion of correctly classified utterances:

$$WA = (TP + TN) / (TP + TN + FP + FN) \quad (11)$$

Because class frequencies are unequal, WA is influenced more by common classes. Unweighted accuracy (UA), equivalently macro-averaged recall, averages per-class recall and so weights each emotion equally:

$$UA = (1/C) \cdot \sum_{c=1}^C TP_c / (TP_c + FN_c) \quad (12)$$

where C is the number of classes. Both are reported throughout, following the convention in this literature. Per-class precision, recall and F1 are additionally reported:

$$\text{Precision} = TP / (TP + FP), \quad \text{Recall} = TP / (TP + FN), \quad F1 = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall}) \quad (13)$$

Differences between configurations are assessed with a two-tailed independent-samples t-test over per-fold measurements, following the procedure used by Rajan et al. (2022), with significance at $\alpha = 0.05$.

4. Materials and Methods

4.1 Hardware and software

Table 2 documents the computational environment. Two platforms were used: a local workstation for development, feature extraction and inference, and a cloud GPU instance for model training, which is approximately twice as fast for the dilated convolutional architecture used here.

Item	Specification
Local processor	Apple M3 Pro, 12 cores, ARM64
Local memory	18 GB unified
Local OS	macOS 14.4 (build 23E214)
Training accelerator	NVIDIA Tesla T4, 16 GB (Google Colab)
Language	Python 3.11.13
Deep learning framework	TensorFlow 2.15.1 with Keras 2 API
Audio processing	librosa 0.11.0 (MFCC, resampling, STFT)
Gammatone features	spafe 0.3.3 (GFCC)
Machine learning	scikit-learn 1.9.0, SciPy 1.17.1
Automatic speech recognition	OpenAI Whisper 20250625, base model
Interface	Gradio 6.26.0
Numerical	NumPy 1.26.4

Table 2. Hardware and software environment.

Framework compatibility. TensorFlow releases from 2.16 onward default to the Keras 3 API, which is incompatible with the Keras 2 code of the reproduced acoustic baseline. Two measures were required: the local environment was pinned to TensorFlow 2.15.1, and on the cloud platform the tf-keras compatibility package was installed with the environment variable `TF_USE_LEGACY_KERAS` set prior to importing TensorFlow. Verification must test the identity of the `tf.keras` module rather than the version of the separately installed keras package, which reports version 3 even when the redirection is active.

4.2 Data acquisition

The IEMOCAP corpus was obtained under licence from the Signal Analysis and Interpretation Laboratory at the University of Southern California. Access requires submission of a release form and acceptance of licence conditions restricting use to non-commercial research, prohibiting redistribution of the data or derivatives, and requiring citation. The corpus is approximately 23 GB; the audio, transcriptions and categorical annotations were used. RAVDESS was obtained from its public Zenodo repository under a Creative Commons Attribution–NonCommercial–ShareAlike 4.0 licence.

4.3 Preprocessing

IEMOCAP distributes emotion labels across per-dialogue annotation files in a bespoke format, transcriptions in separate files, and audio as one file per utterance under a distinct directory hierarchy. A parser was implemented to join the three by utterance identifier, producing a single table with one row

per utterance carrying the audio path, categorical label, original label, speaker, session, temporal boundaries, duration, valence, arousal, dominance and transcript.

Signals were resampled to 22,050 Hz and standardised to a fixed length of 310,000 samples (14.06 s) by symmetric zero-padding of shorter utterances and centre-cropping of longer ones. Fixed length is required by the acoustic encoder. Features were then extracted as described in Section 5.2. Per-feature normalisation used statistics computed on the training partition only.

4.4 Validation of the preprocessing pipeline

The published acoustic baseline distributes its own precomputed features, but these contain only the feature tensor and label array, without utterance identifiers. Transcripts therefore cannot be attached to them, which precludes fusion. An attempt was made to reconstruct their row ordering from the documented file-traversal logic; the resulting alignment correlated with true utterance durations at $r = 0.019$, indistinguishable from random, and was rejected.

Features were therefore regenerated locally using the published extraction settings. Verification compared 25 regenerated utterances against the full published feature set by nearest-neighbour mean absolute difference; all 25 matched to within 0.0001–0.02, consistent with floating-point variation. A subsequent spot check of 12 randomly selected rows from the completed dataset confirmed the same. The regenerated MFCC features are therefore numerically equivalent to the published ones while additionally carrying the identifiers required for fusion.

4.5 Evaluation procedure

All reported results use leave-one-session-out cross-validation, detailed in Section 6.3. Training used the Adam optimiser with learning rate 1×10^{-3} and $\beta_1 = 0.93$, $\beta_2 = 0.98$, batch size 64, and a maximum of 150 epochs with early stopping on validation accuracy at patience 25. Statistical comparison used two-tailed independent-samples t-tests over per-fold measurements.

5. Proposed Methodology and Implementation

5.1 Problem formulation

Let an utterance be represented by a pair (a, w) where a is the acoustic waveform and w the corresponding word sequence. The task is to learn a mapping

$$f: (a, w) \rightarrow \gamma, \quad \gamma \in \{\text{angry}, \text{happy}, \text{neutral}, \text{sad}\} \quad (14)$$

The system decomposes f into three learned components: an acoustic encoder producing a representation of delivery, a semantic encoder producing a representation of content, and a fusion module combining them into a single classification. The fusion module is the object of study; the encoders are held constant across experimental conditions.

5.2 Overall architecture

Figure 1 shows the complete information flow. The acoustic and semantic streams are processed independently up to the fusion stage, at which point both are projected into a shared space and combined by the mechanism under test.

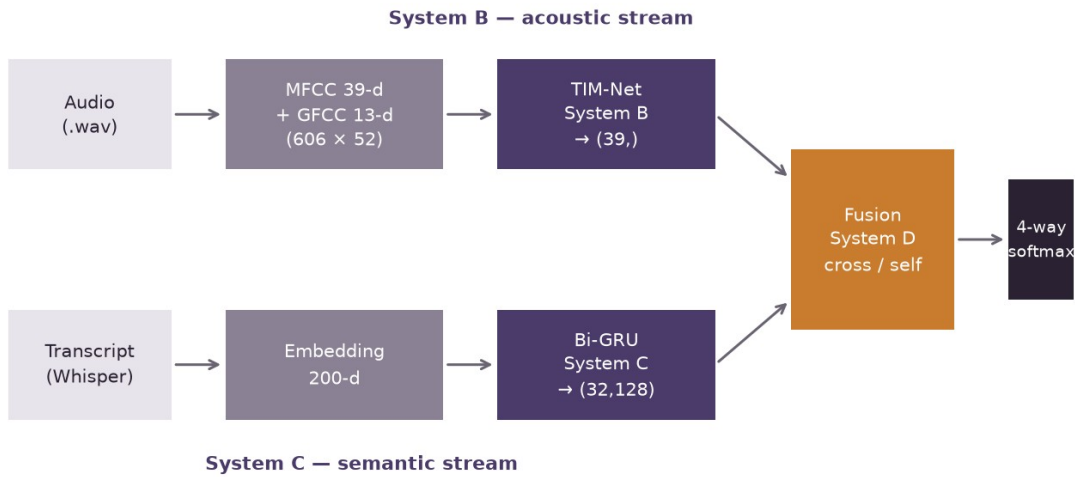


Figure 1. System architecture and information flow. Tensor shapes are given for the MFCC+GFCC configuration.

5.3 Module A — data preparation

Input: the raw IEMOCAP distribution. Processing: annotation, transcript and audio trees are parsed and joined by utterance identifier; labels are mapped to the four-class protocol; signals are standardised to fixed length; features are extracted. Output: an aligned dataset of 5,531 utterances in which row i of every array refers to the same utterance. This alignment is what makes fusion possible and is therefore a functional requirement rather than a convenience.

5.4 Module B — acoustic stream

Input: a standardised waveform of 310,000 samples at 22,050 Hz.

The signal is framed with a 92.9 ms analysis window and 23.2 ms hop, producing 606 frames at 43.1 frames per second. Two cepstral representations are computed per frame: 39 MFCCs from a 128-filter mel filterbank, and 13 GFCCs from a 48-filter gammatone filterbank. The gammatone implementation

yields 602 frames owing to a boundary-handling difference and is edge-padded to 606 for alignment. The two are concatenated along the feature axis to give a 606×52 matrix.

This matrix enters the TIM-Net encoder, which processes it along two paths — forward and time-reversed — through ten Temporal Aware Blocks with dilation rates 1, 2, 4, ..., 512. Each block applies two dilated causal convolutions of kernel size 2, each followed by batch normalisation, ReLU activation and spatial dropout at rate 0.1, then applies a sigmoid to the result and multiplies it elementwise with the block input. The multiplication is significant: the block emits a gate in $[0,1]$ rather than an additive residual, so it learns which frames to propagate rather than merely accumulating features.

By equation (8) the receptive field is $2 \times \sum 2^{(l-1)} = 2,046$ frames, exceeding the 606-frame input, so the deepest blocks condition on the entire utterance. At each of the ten dilation levels the forward and backward paths are summed and average-pooled over time, giving one 39-dimensional vector per level and a 10×39 multi-scale representation. A learned weighting layer of ten scalars reduces this to a single 39-dimensional embedding, allowing the model to select which temporal resolutions are informative.

Output: a 39-dimensional utterance embedding, and a 10×39 sequence made available to the fusion module.

5.5 Module C — semantic stream

Input: the transcript string — ground-truth during training, automatically recognised at inference.

Text is vectorised with lowercasing and punctuation stripping to a fixed length of 32 tokens, embedded into 200 dimensions, and passed through a bidirectional gated recurrent unit with 64 units per direction, producing a 32×128 contextualised sequence together with a padding mask. The vocabulary is limited to 3,000 tokens and is fitted on training-partition text only; fitting it on the full corpus would leak test information through the token inventory.

Output: a 32×128 sequence and its mask.

5.6 Module D — fusion

Input: the acoustic sequence (10×39) and the semantic sequence (32×128) with mask.

Both are linearly projected to a shared 64-dimensional space. Five variants are implemented, of which two are the object of study and three serve as controls:

- Cross-attention. Text queries audio and audio queries text, following equation (9) with Q from one modality and K, V from the other. Both outputs are masked-mean-pooled and concatenated.
- Self-attention. Each modality attends over itself, then both are pooled and concatenated. Structurally identical to the cross variant apart from the origin of Q, K and V .
- Concatenation. Both streams are pooled and concatenated with no attention. A control establishing whether attention contributes at all.
- Audio-only and text-only. Single-stream controls establishing unimodal performance.

Critically, the cross and self variants contain identical numbers of parameters (849,956), because they differ only in the source tensors supplied to the attention operation. Any performance difference is therefore attributable to the mechanism and not to capacity.

Output: a fused vector, passed through dropout, a 64-unit ReLU layer, dropout again, and a four-way softmax.

5.7 Module E — inference interface

Input: an audio recording with no accompanying transcript. Whisper generates a transcript; features are extracted with settings identical to training; the trained model predicts. Output: a predicted emotion, the full probability distribution, the transcript used, and a spectrogram. A comparison mode runs the audio-only, text-only and fusion models on the same input to expose disagreement between streams.

5.8 Algorithm

Algorithm 1: Training and evaluation of the fusion model

```
Input : dataset  $D = \{(a_i, w_i, y_i, \text{spk}_i, \text{sess}_i)\}, i = 1..5531$ 
        fusion mode  $M \in \{\text{cross}, \text{self}, \text{concat}, \text{audio}, \text{text}\}$ 
        feature set  $F \in \{\text{mfcc}, \text{mfcc+gfcc}\}$ 
Output: per-fold weighted and unweighted accuracy; best checkpoint

1   $X \leftarrow \text{ExtractFeatures}(D, F)$  // (5531, 606,  $|F|$ )
2  for  $f \leftarrow 1$  to 5 do // leave-one-session-out
3       $\text{test} \leftarrow \{i : \text{sess}_i = f\}$ 
4       $\text{val} \leftarrow \{i : \text{sess}_i = (f \bmod 5) + 1\}$  // distinct session
5       $\text{train} \leftarrow \text{all indices} \setminus (\text{test} \cup \text{val})$ 
6
7       $V \leftarrow \text{FitVocabulary}(\{w_i : i \in \text{train}\})$  // train text only
8       $\mu, \sigma \leftarrow \text{Mean}(X[\text{train}]), \text{Std}(X[\text{train}])$  // train stats only
9       $X \leftarrow (X - \mu) / \sigma$ 
10
11      $m \leftarrow \text{BuildModel}(M, |F|, V)$ 
12      $\text{best} \leftarrow 0; \text{patience} \leftarrow 0$ 
13     for  $e \leftarrow 1$  to 150 do
14          $\text{TrainOneEpoch}(m, X[\text{train}], w[\text{train}], y[\text{train}])$ 
15          $\text{acc} \leftarrow \text{Evaluate}(m, X[\text{val}], w[\text{val}], y[\text{val}])$ 
16         if  $\text{acc} > \text{best}$  then  $\text{best} \leftarrow \text{acc}; \theta^* \leftarrow \theta; \text{patience} \leftarrow 0$ 
17         else  $\text{patience} \leftarrow \text{patience} + 1$ 
18         if  $\text{patience} \geq 25$  then break // early stopping
19      $\theta \leftarrow \theta^*$  // restore best
20
21      $\hat{y} \leftarrow \text{Predict}(m, X[\text{test}], w[\text{test}])$ 
22      $\text{WA}_f \leftarrow \text{Accuracy}(y[\text{test}], \hat{y}); \text{UA}_f \leftarrow \text{MacroRecall}(y[\text{test}], \hat{y})$ 
23     if  $\text{WA}_f > \text{storedBest}$  then  $\text{SaveCheckpoint}(m, V, \mu, \sigma)$ 
24
25 return  $\text{mean}(\text{WA}), \text{std}(\text{WA}), \text{mean}(\text{UA}), \text{std}(\text{UA})$ 
```

Lines 7 and 8 are the safeguards against leakage: the vocabulary and the normalisation statistics are both derived from the training partition alone. Line 4 ensures the validation session differs from the test session, so early stopping never observes test data — a distinction that is easy to omit and that silently inflates results when omitted.

5.9 Implementation details

- Acoustic encoder: TIM-Net, 39 filters, kernel size 2, one stack, ten dilations, dropout 0.1, ReLU. Used unmodified from the published implementation.
- Semantic encoder: 3,000-token vocabulary, 32-token sequences, 200-dimensional embedding, bidirectional GRU with 64 units per direction, dropout 0.3.
- Fusion: 64-dimensional shared projection, 4 attention heads of 16 dimensions each.
- Classifier: dropout 0.3, dense 64 with ReLU, dropout 0.3, dense 4 with softmax.
- Optimisation: Adam, learning rate 1×10^{-3} , β_1 0.93, β_2 0.98, batch size 64, categorical cross-entropy.

- Stopping: maximum 150 epochs, early stopping on validation accuracy at patience 25 with best-weight restoration.
- Parameter counts: cross-attention 849,956; self-attention 849,956; concatenation 816,676; audio-only 135,430; text-only 673,260.
- Runtime: approximately 300–600 s per fold on the T4 accelerator; the audio-only configuration is the most expensive owing to the depth of the dilated stack.

6. Experimental Framework

6.1 Research questions

The following questions were defined prior to running the experiments reported in Chapter 7.

- RQ1 — Does multimodal fusion outperform each unimodal stream, and by how much?
- RQ2 — Does cross-attention outperform self-attention when the acoustic stream is strengthened, as the confound in Rajan et al. (2022) would predict?
- RQ3 — Does the gammatone extension to the acoustic representation improve performance, and does its effect differ between fusion mechanisms?
- RQ4 — Does any attention mechanism outperform simple concatenation?
- RQ5 — How much of reported accuracy in this task is attributable to evaluation protocol rather than to model capability?
- RQ6 — How well does the trained system generalise to an unseen corpus and to deployment conditions?

6.2 Dataset description

Table 3 documents the primary corpus in the detail required to assess composition and potential bias.

Item	Description
Source	IEMOCAP, USC SAIL; obtained under licence by application
Samples	5,531 utterances retained from 10,039 recordings
Classes	angry 1,103; happy 1,636; neutral 1,708; sad 1,084
Acquisition	Ten actors, five two-person sessions; scripted (2,588) and improvised (2,943)
Audio	6.99 h total; mean 4.55 s, range 0.58–34.14 s per utterance
Labels	Categorical, by majority vote of three annotators; also valence, arousal, dominance on a 1–5 scale
Splits	Leave-one-session-out; per fold approximately 3,300 train, 1,000 validation, 1,000 test
Imbalance	Largest class 30.9%, smallest 19.6%; no resampling applied, unweighted accuracy reported alongside weighted
Augmentation	None
Exclusions	Non-consensus labels (2,507), frustrated (1,849), and classes with fewer than 200 examples (152)

Table 3. Dataset description.

Table 4 details the label mapping and the exclusions, which follow the standard four-class protocol in this literature.

Original	Mapped to	Count	Rationale
ang	angry	1,103	Retained
hap	happy	595	Retained
exc	happy	1,041	Merged with happy, following convention in this literature

Original	Mapped to	Count	Rationale
neu	neutral	1,708	Retained
sad	sad	1,084	Retained
fru	excluded	1,849	Frustrated; acoustically overlapping with angry and neutral
xxx	excluded	2,507	No annotator consensus
sur, fea, oth, dis	excluded	152	Insufficient examples for reliable learning

Table 4. Label mapping and exclusions. The retained subset totals 5,531 utterances.

Table 5 shows per-session composition. Because each session contains two speakers who appear in no other session, holding out a session holds out two unseen voices — the property that makes the protocol genuinely speaker-independent.

Session	Speakers	Utterances	Angry	Happy	Neutral	Sad
1	Ses01F, Ses01M	1,085	229	278	384	194
2	Ses02F, Ses02M	1,023	137	327	362	197
3	Ses03F, Ses03M	1,151	240	286	320	305
4	Ses04F, Ses04M	1,031	327	303	258	143
5	Ses05F, Ses05M	1,241	170	442	384	245

Table 5. Per-session composition of the corpus.

6.3 Split strategy

Table 6 gives the fold structure. Each fold designates one session as test, a different session as validation, and the remaining three as training.

Fold	Test	Validation	Training	Train size
1	Session 1	Session 2	Sessions 3, 4, 5	3,423
2	Session 2	Session 3	Sessions 1, 4, 5	3,357
3	Session 3	Session 4	Sessions 1, 2, 5	3,349
4	Session 4	Session 5	Sessions 1, 2, 3	3,259
5	Session 5	Session 1	Sessions 2, 3, 4	3,205

Table 6. Cross-validation fold structure. Validation and test sessions are always distinct, so early stopping never observes test data.

6.4 Selection of the primary corpus

Initial development used RAVDESS, on which the acoustic baseline was first reproduced (92.08% against 90.04/90.07 published). An evaluation audit conducted before extending the system, however, established that RAVDESS is structurally unsuited to the speaker-independent claims this project intended to make.

RAVDESS records every experimental condition exactly twice. An inventory of the corpus confirmed that all 720 conditions have a second take, so every clip has a near-duplicate counterpart. Under randomised cross-validation, one take is assigned to training and the other to test, and a model can score highly by recognising a recording it has effectively already seen. The magnitude of the effect is quantified in Section 7.5.

Change of primary corpus. For this reason RAVDESS was retained only as a diagnostic corpus for the leakage study and for cross-corpus testing, and IEMOCAP was adopted as the primary evaluation corpus. IEMOCAP consists of spontaneous and scripted dyadic dialogue with no systematic retakes, and its session structure supports genuinely speaker-independent partitioning. The same audit applied to IEMOCAP measured an inflation of 7.96 points, approximately one third that observed on RAVDESS, confirming the corpus is materially more suitable.

6.5 Baselines

Five baselines are used, all trained and evaluated under the identical protocol described above. Audio-only and text-only establish unimodal performance. Concatenation establishes whether attention contributes beyond simply presenting both streams. Self-attention is the stronger of the two mechanisms reported by Rajan et al. (2022). Published results for the acoustic encoder are additionally reproduced, and are reported as reproductions with their original protocol explicitly labelled, since that protocol differs from the one used here.

6.6 Metrics and statistical procedure

Weighted and unweighted accuracy are reported for all configurations as mean $\pm$ standard deviation across the five folds, with per-class precision, recall and F1 for the principal model. Comparisons between configurations use a two-tailed independent-samples t-test over per-fold values at $\alpha = 0.05$. Reporting both WA and UA is necessary because the corpus is imbalanced; a configuration achieving high WA through the majority class alone would be identifiable by a divergence between the two.

7. Results and Discussion

7.1 Baseline reproduction

Before any extension, the published acoustic baseline was reproduced to establish that the environment, features and evaluation code were correct. Under the original authors' own protocol, RAVDESS yielded 92.08% against a published 90.04/90.07, and IEMOCAP yielded 71.65% against published figures in the range 68–72%. Both reproduce.

The RAVDESS figure slightly exceeds the published value because the evaluation mode loads per-fold checkpoints that were selected for peak accuracy, which is optimistic by construction. This verifies the published weights rather than constituting an independent retraining, and is reported as such.

7.2 Primary result: fusion against unimodal streams (RQ1)

Table 7 compares all configurations under the identical leave-one-session-out protocol. The first two rows establish unimodal performance; the remaining rows are fusion variants.

Configuration	Features	WA (%)	UA (%)
Audio only (System B)	MFCC	53.86 $\pm$ 1.62	55.56 $\pm$ 1.59
Text only (System C)	MFCC	60.15 $\pm$ 2.89	61.15 $\pm$ 3.17
Concatenation, no attention	MFCC+GFCC	67.65 $\pm$ 3.43	69.44 $\pm$ 3.62
Cross-attention	MFCC	66.06 $\pm$ 3.10	67.67 $\pm$ 3.54
Cross-attention	MFCC+GFCC	67.22 $\pm$ 3.75	68.66 $\pm$ 4.31
Self-attention	MFCC	66.08 $\pm$ 2.56	67.43 $\pm$ 2.78
Self-attention	MFCC+GFCC	66.28 $\pm$ 2.92	67.83 $\pm$ 4.05

Table 7. Primary results across all configurations, mean $\pm$ standard deviation over five folds. Chance is 25%.

Fusion exceeds the stronger unimodal stream by 7.50 points and the weaker by 13.79 points. This margin is large relative to the fold-to-fold standard deviation and constitutes the most robust finding in this work: the two modalities carry genuinely complementary information, and combining them produces reliable improvement. It also answers, on measured evidence, the question of whether a speech-only system would have sufficed — audio alone is the weakest configuration evaluated.

A secondary observation concerns modality ordering. Text alone (60.15%) substantially outperforms audio alone (53.86%), reproducing the ordering reported by Rajan et al. (2022), who found text at 0.474 against audio at 0.365. That two independent implementations on different task formulations recover the same ordering indicates it is a property of the task rather than of a particular architecture.

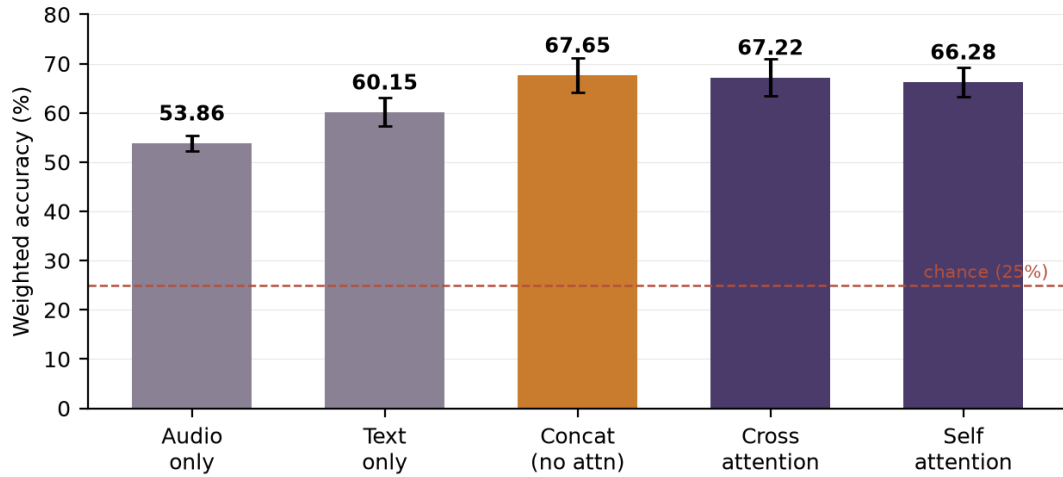


Figure 2. Weighted accuracy by configuration, with standard deviation across folds. The three fusion configurations are statistically indistinguishable from one another; all three exceed both unimodal streams.

7.3 Cross-attention against self-attention (RQ2)

Table 8 presents the central comparison. Because the two variants have identical parameter counts and are trained on identical data with identical hyperparameters, any difference is attributable to the attention mechanism alone.

Features	Metric	Cross	Self	Difference	p-value
MFCC	WA	66.06	66.08	-0.02	0.991
MFCC	UA	67.67	67.43	+0.24	0.919
MFCC+GFCC	WA	67.22	66.28	+0.94	0.702
MFCC+GFCC	UA	68.66	67.83	+0.83	0.786

Table 8. Cross-attention versus self-attention, two-tailed independent-samples t-test over per-fold values.

No configuration produces a significant difference. The smallest p-value observed is 0.702, far above the 0.05 threshold, and the largest difference is 0.94 points — well within the fold-to-fold standard deviation of approximately 3 points. The two mechanisms are statistically comparable.

The hypothesis was not supported. This project predicted that strengthening the acoustic stream would allow cross-attention to outperform self-attention, on the reasoning that its failure in prior work was attributable to a weak audio branch. The acoustic stream was strengthened substantially — the encoder reproduces 71.65% in isolation, against 36.5% for the audio branch in the study that motivated the hypothesis — and cross-attention still did not win. The direction of the effect is weakly consistent with the hypothesis, in that cross-attention moves from level with self-attention on MFCC alone to 0.94 points ahead once gammatone features are added, but at $p = 0.702$ this is indistinguishable from noise and cannot be claimed as evidence.

The result does, however, constitute an independent replication of the published null under materially different conditions: four emotion classes rather than seven, a dilated temporal convolutional encoder rather than a convolutional-recurrent one, an extended feature set, and a different implementation framework. Replication under changed conditions is stronger evidence than a direct re-run, and it suggests the null is a property of the fusion problem rather than an artefact of one experimental setup.

7.4 Ablation (RQ3, RQ4)

Table 9 isolates the contribution of each component by removing or substituting one at a time.

Configuration	Audio	Text	Attention	WA (%)	Δ vs full
Audio only	Yes	No	None	53.86	-13.79
Text only	No	Yes	None	60.15	-7.50
Concatenation	Yes	Yes	None	67.65	reference
Self-attention	Yes	Yes	Self	66.28	-1.37
Cross-attention	Yes	Yes	Cross	67.22	-0.43

Table 9. Ablation study. All configurations use MFCC+GFCC except the two unimodal controls; the concatenation configuration is taken as reference because it is the highest-scoring fusion variant.

Two conclusions follow. First, the dominant contribution is the presence of the second modality, worth between 7.5 and 13.8 points depending on which stream is removed. Second, and unexpectedly, the attention mechanism contributes nothing measurable: plain concatenation attains the highest weighted accuracy of any fusion variant, exceeding cross-attention by 0.43 points and self-attention by 1.37. With a single seed these differences lie within the noise, so the defensible claim is that cross-attention, self-attention and no attention at all are statistically indistinguishable on this task.

This is a stronger statement than the published finding it replicates. Rajan et al. (2022) established that cross-attention is comparable to self-attention; the present ablation indicates that both are comparable to using no attention mechanism whatsoever. Given that attention adds approximately 33,000 parameters and additional computation, this has practical implications for deployment as well as architectural ones.

Regarding the gammatone extension, adding GFCC improves cross-attention by 1.16 points ($p = 0.645$) and self-attention by 0.20 points ($p = 0.921$). Neither approaches significance. The extension costs 1,014 parameters, an increase of 0.76%, so it is inexpensive — but on this evidence it cannot be claimed to help.

7.5 Evaluation-protocol leakage (RQ5)

This analysis was not part of the original plan. It began as a consistency check on an unexpectedly large gap between reproduced and expected accuracies, and became the most transferable finding of the project. In each comparison below, features, model architecture and hyperparameters are held constant and only the train–test partitioning is varied.

Corpus and model	Random split	Speaker-independent	Inflation
RAVDESS, mean-pooled SVM	70.49	50.83	+19.65
RAVDESS, 1D CNN	86.31	65.83	+20.47
IEMOCAP, mean-pooled SVM	64.06	56.09	+7.96

Table 10. Evaluation-protocol leakage. Only the partitioning differs between the two columns.

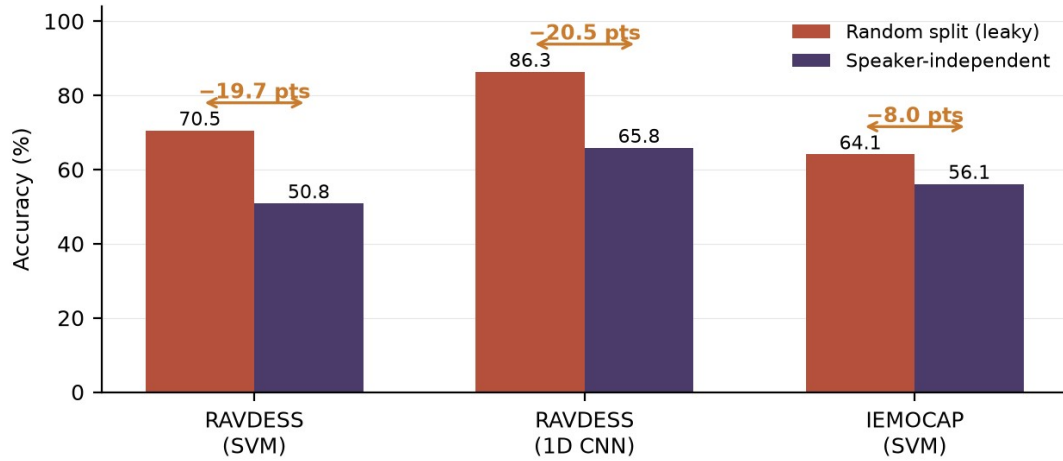


Figure 3. Evaluation-protocol inflation across corpora and models. The effect is approximately 20 points on RAVDESS and 8 points on IEMOCAP.

The initial hypothesis was that speaker overlap caused the inflation. Decomposition disproved this. Blocking near-duplicate takes while permitting speaker overlap reduced accuracy by 21.32 points; additionally blocking speaker overlap changed the result by -1.67 points, within noise. The cause is therefore duplicate recordings, not speaker identity.

Table 11 quantifies the mechanism by measuring acoustic similarity between categories of clip pair.

Clip pair	Mean cosine similarity
Same actor, emotion and sentence — take 1 against take 2	0.600
Same actor, different emotion	0.197
Same emotion, different actor	0.128
Two randomly selected clips	0.014

Table 11. Acoustic similarity between clip pairs in RAVDESS, mean cosine similarity in the pooled feature space.

A clip resembles its own second take approximately five times more strongly than it resembles another speaker expressing the same emotion. In this representation, recordings cluster by recording rather than by emotion, which explains how a randomised split permits a model to reach 86.31% by memorisation. A corroborating measurement reinforces this: the same features predict which of the 24 actors is speaking at 95.28% accuracy against 70.49% for emotion, indicating that the representation encodes speaker identity more strongly than affect.

Scope of this finding. This result is corpus-specific and should not be generalised into a claim that published accuracies in this field are broadly unreliable. IEMOCAP's community-standard leave-one-session-out protocol is honest: it measured 55.27% against 56.09% for speaker-independent cross-validation, a difference within noise. The problem belongs to acted corpora containing systematic retakes, of which RAVDESS is the clearest case, and it is the reason RAVDESS was not adopted as the primary corpus for this project.

7.6 Cross-corpus generalisation (RQ6)

The system trained on IEMOCAP was evaluated without adaptation on 60 randomly sampled RAVDESS clips restricted to the four shared classes, with the correct carrier sentence supplied as transcript so that recognition error could not confound the measurement.

Evaluation condition	WA (%)	Relative to chance
Held-out IEMOCAP session	69.79	+44.79
RAVDESS, no adaptation	35.00	+10.00
Chance (four classes)	25.00	—

Table 12. Cross-corpus generalisation of the IEMOCAP-trained fusion model.

Performance falls to 35.0%, only 10 points above chance, and the error structure is informative: of 21 sad clips, 9 were classified as happy, and the model defaults to the happy class across most inputs. The degradation is consistent with the cross-corpus generalisation problem widely reported in this literature and attributable to differences in recording chain, microphone characteristics and acting style. It establishes a concrete boundary on the claims made here: the reported accuracies apply to IEMOCAP-like speech and should not be assumed to transfer.

7.7 Per-class performance and error analysis

Table 13 gives per-class metrics for the fusion model on its held-out session, and Figure 4 the corresponding confusion matrix.

Class	Precision	Recall	F1	Support
Angry	0.615	0.920	0.737	137
Happy	0.661	0.780	0.715	327
Neutral	0.728	0.481	0.579	362
Sad	0.782	0.766	0.774	197
Macro average	0.696	0.737	0.701	1,023

Table 13. Per-class performance of the fusion model, held-out session 2, $n = 1,023$.

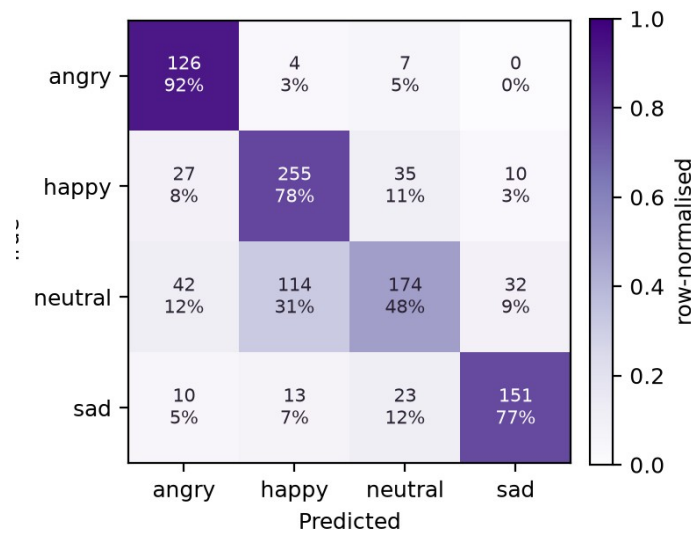


Figure 4. Confusion matrix of the fusion model on the held-out session, annotated with counts and row-normalised percentages.

The dominant error mode is neutral, which attains recall of only 0.481 despite being the largest class. Of 362 neutral utterances, 114 are classified as happy and 42 as angry. Angry shows the opposite profile —

recall 0.920 but precision 0.615 — indicating the model over-predicts angry, drawing false positives principally from neutral.

This is interpretable. Neutral is not an emotion with positive acoustic markers but the absence of strong markers, which makes it a residual category rather than a distinct one. Any utterance with moderate energy or slight pitch variation can be pulled toward an expressive class. The asymmetry between angry recall and precision indicates the decision boundary between neutral and the expressive classes sits too close to neutral, and class-weighted loss or a calibrated threshold would be the natural remedy.

The divergence between weighted accuracy (0.690) and macro-averaged recall (0.737) is modest, indicating the model is not simply exploiting the majority class.

7.8 Deployment characteristics

The inference interface was evaluated on previously unseen audio. Feature extraction and prediction complete in approximately 1.5 s per utterance once the recognition model is resident in memory; the first invocation incurs approximately 30 s of model loading. Transcription quality on corpus audio was high — for example, ground truth "Well what's the problem? Let me change it." was recognised as "Well, what's the problem? Let me change it."

A qualitative evaluation on twelve held-out clips found the three streams disagreeing on eleven, including cases in which both unimodal streams were incorrect and fusion was nevertheless correct. One representative case is an utterance with the transcript "Look at you. Look at you shaking.", whose true label is angry: the text stream classifies it as happy with confidence 0.481, since the lexical content is superficially complimentary, while the acoustic stream identifies angry at 0.955 and fusion follows at 0.563. This is precisely the ambiguity case motivating the project, and it demonstrates the mechanism operating as intended on data the model has not seen.

Benchmark accuracy is not deployment accuracy. Three factors separate the reported 66.28% from live performance. Training used ground-truth transcripts whereas deployment uses automatic recognition; microphone and room characteristics differ from the recording conditions of the corpus; and the label space contains only four emotions, so any other expressed state is necessarily misclassified. Informal testing on laptop-microphone recordings produced markedly worse results than corpus audio, consistent with the cross-corpus measurement in Section 7.6.

8. Conclusion and Future Work

8.1 Conclusion

This project addressed whether cross-modal attention — the prevailing fusion mechanism in multimodal emotion recognition — confers a measurable advantage over self-attention when the acoustic stream is strengthened, a question left unresolved by a confound in the study that first reported the comparison.

A dual-stream system was developed combining a reproduced temporal convolutional acoustic encoder, extended with gammatone cepstral coefficients, and a recurrent semantic encoder, with a fusion module implementing both attention mechanisms at identical parameter count. All evaluation used leave-one-session-out cross-validation on IEMOCAP.

The principal findings are as follows. Multimodal fusion substantially outperformed both unimodal streams, reaching 67.65% weighted accuracy against 60.15% for text and 53.86% for audio; this is the most reliable result obtained. The central hypothesis, however, was not supported: cross-attention reached 67.22% against 66.28% for self-attention, a difference of 0.94 points at $p = 0.702$, and the gammatone extension produced no significant improvement for either mechanism. Neither attention variant exceeded plain concatenation at 67.65%, indicating that on this task the attention machinery does not justify its complexity. Separately, an evaluation audit measured accuracy inflation of 19.65 to 20.47 points on RAVDESS and 7.96 points on IEMOCAP attributable to partitioning strategy alone, with the RAVDESS effect traced to duplicate recordings rather than speaker overlap — a finding that determined the choice of primary corpus for this work.

The implication is that, for audio–text emotion recognition, the modality composition and the evaluation protocol exert greater influence on measured performance than the choice of fusion mechanism. Attention is widely adopted as a default; on this evidence that default is not earning its cost.

The principal limitation is that the attention comparisons rest on a single random seed of five folds. Given p -values between 0.702 and 0.991, additional seeds are unlikely to reverse the conclusion, but they are required before the claim is fully secured. A secondary limitation is that all results are obtained on a single corpus, and Section 7.6 demonstrates these do not transfer to an unseen one.

8.2 Future work

- Repeat the attention comparison across additional random seeds to narrow the confidence intervals on the null result, and extend it to a second corpus so the replication is not corpus-bound.
- Address the neutral-class failure identified in Section 7.7, where recall of 0.481 drives most residual error. Class-weighted loss, threshold calibration, or explicit modelling of neutral as a residual rather than a positive category are the direct candidates.
- Quantify the automatic-recognition penalty by evaluating the fusion model with transcripts degraded to controlled word-error rates, converting the qualitative deployment observation in Section 7.8 into a measured curve.
- Apply the leakage decomposition of Section 7.5 to the published acoustic baseline under its own protocol, which would isolate how much of its reported 71.65% on IEMOCAP is attributable to partitioning rather than architecture — a measurement not present in the literature.
- Evaluate whether a self-supervised speech representation trained on large unlabelled corpora closes the cross-corpus gap of Section 7.6, where the present system retains only 35.0% accuracy.

- Extend the label space beyond four classes, since the current formulation guarantees misclassification of any state outside it, and evaluate on spontaneous rather than acted speech to test ecological validity.

References

- Busso, C., Bulut, M., Lee, C.-C., Kazemzadeh, A., Mower, E., Kim, S., Chang, J. N., Lee, S., & Narayanan, S. S. (2008). IEMOCAP: Interactive emotional dyadic motion capture database. *Language Resources and Evaluation*, 42(4), 335–359.
- Dutta, S., & Ganapathy, S. (2023). HCAM — Hierarchical cross attention model for multi-modal emotion recognition. *arXiv:2304.06910*.
- Hazarika, D., Zimmermann, R., & Poria, S. (2020). MISA: Modality-invariant and -specific representations for multimodal sentiment analysis. *Proceedings of the 28th ACM International Conference on Multimedia*, 1122–1131.
- Li, F., Wu, K., & Li, Y. (2025). MCN-CL: Multimodal cross-attention network and contrastive learning for multimodal emotion recognition. *arXiv:2511.10892*.
- Livingstone, S. R., & Russo, F. A. (2018). The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS): A dynamic, multimodal set of facial and vocal expressions in North American English. *PLOS ONE*, 13(5), e0196391.
- Poria, S., Hazarika, D., Majumder, N., Naik, G., Cambria, E., & Mihalcea, R. (2019). MELD: A multimodal multi-party dataset for emotion recognition in conversations. *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 527–536.
- Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2023). Robust speech recognition via large-scale weak supervision. *Proceedings of the 40th International Conference on Machine Learning*, 28492–28518.
- Rajan, V., Brutti, A., & Cavallaro, A. (2022). Is cross-attention preferable to self-attention for multi-modal emotion recognition? *ICASSP 2022 — IEEE International Conference on Acoustics, Speech and Signal Processing*, 4693–4697.
- Tsai, Y.-H. H., Bai, S., Liang, P. P., Kolter, J. Z., Morency, L.-P., & Salakhutdinov, R. (2019). Multimodal Transformer for unaligned multimodal language sequences. *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 6558–6569.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30, 5998–6008.
- Ye, J., Wen, X.-C., Wei, Y., Xu, Y., Liu, K., & Shan, H. (2023). Temporal modeling matters: A novel temporal emotional modeling approach for speech emotion recognition. *ICASSP 2023 — IEEE International Conference on Acoustics, Speech and Signal Processing*, 1–5.
- Zadeh, A., Chen, M., Poria, S., Cambria, E., & Morency, L.-P. (2017). Tensor Fusion Network for multimodal sentiment analysis. *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, 1103–1114.